

Pranjal Narwal

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Summary

Full-Stack Engineer who single-handedly builds production-grade systems that typically take a team of five — a real-time collaborative spreadsheet with live multi-user sync, a distributed chaos-engineering simulator modeling cascading failures, and an AI-driven travel platform spanning 40+ countries — each architected and shipped solo, end-to-end from schema to deployment, alongside 200+ solved LeetCode problems, as an undergraduate.

Education

Manipal University Jaipur

Aug 2023 – May 2027 (Expected)

B.Tech — Computer Science and Engineering - Jaipur, Rajasthan

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

Experience

Coradius Solutions

Jun 2026 – Jul 2026

Junior Software Developer — Intern - Agra, Uttar Pradesh

- Developed **attendance and fee management modules** for Accretion+, a production School/College ERP platform used by **100+ schools and businesses** nationwide, using React, Node.js, and MySQL.
- Implemented **role-based access control (RBAC)** across administrator, teacher, and student user types to enforce differentiated permissions and secure data access.
- Designed **relational database schemas** modeling real-world entities — student records, class hierarchies, and fee structures — translating academic DBMS concepts into production schema design.

LaunchEd Global EdTech

Dec 2025 – May 2026

Brand Executive — Campus Ambassador - Bengaluru, Karnataka (Remote)

- Drove **student registrations** for internship programmes over a **6-month engagement** by designing targeted digital campaigns across social media platforms using data-driven outreach strategies.
- Managed **end-to-end campus campaign operations**, coordinating between the organisation and the student community to consistently hit performance milestones.

Projects

Dora Sheet

TypeScript · React 18 · NestJS · Socket.IO · PostgreSQL · Redis · Docker · Vercel - [Live Demo](#) - [GitHub](#)

- Shipped a production-grade collaborative spreadsheet with **simultaneous multi-user editing**, live cursor tracking, Web-Socket sync, and **invitation-only rooms** secured by 6-digit access codes.
- Built a proactive AI assistant using **structured multi-tool function calling** with a **9-step decision hierarchy**, enabling natural-language spreadsheet manipulation with a safety-first Approve & Apply workflow.
- Delivered a complete spreadsheet engine with **4 core formula functions** (SUM, AVERAGE, IF, VLOOKUP), **Excel (.xlsx) import** via SheetJS, undo/redo history, version snapshots, and find-and-replace.
- Designed a scalable monorepo with NestJS + **JWT auth**, Prisma + PostgreSQL, **Redis** for distributed cell-level locking, Docker Compose, and a **GitHub Actions CI/CD** pipeline across Vercel & Render.

Archaos — Distributed Systems Chaos Engineering Simulator

TypeScript · React 19 · NestJS · PostgreSQL · Redis · Web Worker · Docker · Vercel · Railway - [Live Demo](#) - [GitHub](#)

- Built a **discrete-event simulation engine** on an off-thread Web Worker using a min-heap priority queue ($O(\log n)$ event ordering), sustaining **60 fps** canvas rendering at **250ms ticks** across **12 node types** — zero UI-thread blocking.
- Engineered dual finite-state machines — a **5-state node health FSM** (HEALTHY → DEGRADED → FAILING → DEAD → RECOVERING) and a per-edge circuit breaker FSM (CLOSED → OPEN → HALF-OPEN) — modeling cascading failures, retry-storm amplification, and OOM restart cycles.
- Shipped **8 chaos injection types**, **8 war-game scenarios** (Split Brain, Thundering Herd, Queue Flood), a **3-format IaC parser** (Docker Compose · Kubernetes YAML · Terraform HCL), and a **13-endpoint headless REST API + CLI** with SLA-compliance exit codes for CI/CD pipeline integration.
- Streamed real-time AI narration via OpenRouter with automatic model fallback; computed blast-radius propagation via recursive PostgreSQL CTE BFS; secured with RS256 JWT, bcrypt, and Prisma parameterised queries.

ShadowCart

TypeScript · React 19 · Vite · CRXJS · IndexedDB · Brevo API · Vercel Serverless - [GitHub](#)

- Built a Chrome extension intercepting Add-to-Cart events across **major e-commerce sites** (Amazon, Shopify, luxury retailers), maintaining a private unified wishlist with **zero account creation required**.
- Implemented a **Cooling Period reminder system** with configurable delay timers and automated email notifications via Brevo API — reducing impulsive purchases through behavioural design.
- Adopted a **local-first, privacy-preserving architecture** using IndexedDB — **zero third-party data transmission** and complete user ownership of all shopping history.
- Packaged via Vite + CRXJS for Chrome deployment; Vercel Serverless Functions handle secure email logic and behavioural spending insights.

Dora Travel Agency

Python · Flask · JavaScript · Three.js · Leaflet.js · REST APIs · Vercel - [Live Demo](#) - [GitHub](#)

- Built a production-ready AI platform covering **40+ countries**, generating structured multi-day itineraries, cultural advisories, and destination briefs from a single user query.
- Engineered an interactive **3D globe** (Three.js + Globe.gl) with animated flight arcs and zoom-to-destination transitions, significantly boosting user engagement and visual storytelling.
- Architected a Flask REST API integrating **5 external data sources** (Open-Meteo, Overpass/OSM, Frankfurter) across **11 real-time intelligence widgets** and **9 curated intelligence databases** — currency converter, SIM guide, scam advisories, and daily budget estimator.
- Delivered a **multilingual UI in 4 languages** (EN, HI, ES, FR) with a **3-tier dynamic pricing engine**, deployed serverlessly with **zero-downtime** availability.

Technical Skills

- **Languages:** C, C++, Java, Python, TypeScript, JavaScript (ES6+), HTML5, CSS3, SQL
- **Frontend:** React 18, React 19, Vite, Tailwind CSS, Three.js, Leaflet.js, Globe.gl, Zustand, CRXJS
- **Backend:** Node.js, NestJS, Flask, REST API Design, Socket.IO, WebSockets, Serverless Functions
- **Databases:** PostgreSQL, Redis, IndexedDB, Prisma ORM
- **Cloud and DevOps:** Vercel, Render, Git, GitHub, GitHub Actions, CI/CD, Postman, VS Code
- **Core Concepts:** Data Structures and Algorithms, System Design, Real-Time Systems, AI Function Calling, Browser Extension Development, Role-Based Access Control, Relational Database Design, API Architecture, Authentication, JWT, Microservices Concepts, Version Control

Achievements & Recognition

- **International Innovation Challenge 2.0** — Cleared Round 1 and advanced to Round 2 out of a large national pool of participants across India.
- **200+ LeetCode Challenges** — Consistent DSA practice across Arrays, Trees, Graphs, Dynamic Programming, and Sliding Window techniques.
- **4 Production Apps Shipped** — Independently architected and deployed multiple complex, AI-integrated full-stack applications as a 4-year undergraduate.